Keshav Sharma

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EDUCATION

Vellore Institute of Technology — **Chennai, Tamil Nadu**

Bachelor of Technology in Computer Science (Data Science Specialisation) — *Graduation: May 2027*

SKILL SUMMARY

Programming Languages: Python, SQL, R, Java
Data Science & Machine Learning: Pandas, NumPy, Scikit-learn, Statistical Analysis, Exploratory Data Analysis (EDA), Feature Engineering, Predictive Modeling, Anomaly Detection, Model Evaluation
Machine Learning Techniques: BG/NBD, Gamma-Gamma Models, Random Forest, Decision Trees, Naive Bayes, Support Vector Machines (SVM), Time-Series Analysis
Generative AI: LLMs, Retrieval-Augmented Generation (RAG), Prompt Engineering, Google Gemini API, Semantic Search
Databases & Data Engineering: SQLite, MySQL, MongoDB, SQL Querying, Data Cleaning, Data Preprocessing, Data Modeling, Data Validation, Data Pipelines
Analytics & Visualization: Power BI, DAX, Tableau, Matplotlib, Seaborn, Business Intelligence, Customer Segmentation

EXPERIENCE

AI for Sustainability Virtual Intern* |LINK** — ***July – Sep, 2026

- Completed an AI-focused virtual internship covering Generative AI, Prompt Engineering, RAG, Agentic AI, and Responsible AI, with hands-on exposure to enterprise AI workflows and LLM applications.
- Applied RAG and semantic search concepts using document ingestion, embeddings, vector databases, and retrieval pipelines to improve the grounding and reliability of LLM responses.
- Built an Energy Anomaly Detector using Python, Pandas, NumPy, and Isolation Forest, detecting 9 anomalies across 6,918 test hours (0.13%), with automated email alerts, privacy safeguards, transparent scoring, and rate-limited notifications.

PROJECTS

***Financial Transaction Anomaly Detection & Audit Dashboard* | LINK**

- Engineered an end-to-end Python analytics pipeline processing 50K+ PaySim transactions, incorporating data validation, duplicate detection, statistical outlier analysis, vendor-velocity checks, HHI concentration analysis, and Isolation Forest.
- Designed a star-schema SQLite data warehouse and implemented a composite 0–100 risk-scoring framework, identifying 357 high/critical-risk transactions with end-to-end analysis completed in under 7 seconds.
- Developed a 5-page Power BI dashboard with 20+ DAX measures to deliver vendor risk, anomaly-rate, and transaction trend insights, supported by automated pipeline tests..

***Docwise* |LINK**

- Built an AI-powered PDF question-answering system using Google Gemini and RAG, enabling contextual question answering over documents exceeding 200 pages.
- Developed a document processing and retrieval pipeline combining PDF extraction, chunking, embeddings, and semantic retrieval, achieving 95%+ query accuracy.
- Integrated FastAPI and LLM inference to deliver an end-to-end AI application for document understanding and conversational querying.

***Customer Lifetime Value Engine* |LINK**

- Built probabilistic BG/NBD and Gamma-Gamma models on 1M+ historical transaction records to forecast customer purchasing behaviour and lifetime value.
- Segmented customers into value-based cohorts, identifying the top 20% of customers contributing ~65% of forecasted revenue to support business prioritisation.
- Performed transaction-level preprocessing, frequency/recency analysis, and model validation; visualised customer value insights through Power BI dashboards for stakeholder reporting.

***Real-Time Stock Market Anomaly Detector* |LINK**

- Engineered a real-time market analytics pipeline processing 2M+ NSE records, combining price signals and financial-news sentiment to detect unusual market behaviour.
- Developed and evaluated Isolation Forest and LSTM Autoencoder models, achieving 88% precision and 85% recall on validation data for anomaly detection.
- Optimised the analytics pipeline to reduce detection latency by 35%, enabling near real-time monitoring and faster identification of unusual market events.

Certifications

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional |LINK — 2026
- Deloitte Australia Data Analytics Job Simulation |LINK — 2025